



Question 1

(a) $x(t) = t^3 \sin(2t)u(t)$

Using the differentiation in frequency domain property:

$$t^n f(t) \xleftrightarrow{\mathcal{L}} (-1)^n \frac{d^n}{ds^n} F(s)$$

For the base signal $f(t) = \sin(2t)u(t)$:

$$F(s) = \frac{2}{s^2 + 4}$$

Applying the property for $n = 3$:

$$X(s) = -\frac{d^3}{ds^3} \left(\frac{2}{s^2 + 4} \right) = \frac{48s(s^2 - 4)}{(s^2 + 4)^4} ; \quad \text{ROC: } \text{Re}\{s\} > 0$$

Initial Value:

$$\lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s) = 0$$

Final Value: Since it diverges if t goes to infinity so it doesn't have initial value

(b) $x(t) = e^{-2|t|} \sin(2t)$

Since this signal has values for $t < 0$ so, it is a non-casual signal and initial value is not valid, also for region of convergence:

$$x(t) = e^{-2t} \sin(2t)u(t) + e^{2t} \sin(2t)u(-t)$$

Right-sided component:

$$\mathcal{L}\{e^{-at} \sin(\omega_0 t)u(t)\} = \frac{\omega_0}{(s + a)^2 + \omega_0^2} ; \quad \text{ROC: } \text{Re}\{s\} > -a$$

Substituting $a = 2, \omega_0 = 2$:

$$\mathcal{L}\{e^{-2t} \sin(2t)u(t)\} = \frac{2}{(s + 2)^2 + 4} ; \quad \text{ROC: } \text{Re}\{s\} > -2$$

Left-sided component:

$$\mathcal{L}\{e^{at} \sin(\omega_0 t)u(-t)\} = \frac{\omega_0}{(s - a)^2 + \omega_0^2} ; \quad \text{ROC: } \text{Re}\{s\} < a$$

Substituting $a = 2, \omega_0 = 2$:

$$\mathcal{L}\{e^{2t} \sin(2t)u(-t)\} = \frac{2}{(s - 2)^2 + 4} ; \quad \text{ROC: } \text{Re}\{s\} < 2$$

Total Transform:

$$X(s) = \frac{2}{(s + 2)^2 + 4} - \frac{2}{(s - 2)^2 + 4} ; \quad \text{ROC: } -2 < \text{Re}\{s\} < 2$$

the ROC is not right handed so the final value is not valid too.



(c) $x(t) = \frac{\sin(t)}{t}u(t)$

Using the integration in frequency domain property:

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty F(\alpha) d\alpha$$

For $f(t) = \sin(t)u(t)$, we have $F(s) = \frac{1}{s^2+1}$. Thus:

$$\mathcal{L}\{x(t)\} = \int_s^\infty \frac{1}{\alpha^2+1} d\alpha$$

Evaluating the integral:

$$X(s) = [\tan^{-1}(\alpha)]_s^\infty = \frac{\pi}{2} - \tan^{-1}(s) \quad ; \quad \text{ROC: } \text{Re}\{s\} > 0$$

Initial and Final Values:

$$\lim_{t \rightarrow 0} x(t) = \lim_{s \rightarrow \infty} sX(s) = 0 \quad ; \quad \lim_{t \rightarrow \infty} x(t) = \lim_{s \rightarrow 0} sX(s) = 0$$

(d) $x(t) = |t-1|e^{-|t|}$

$$X(s) = \int_{-\infty}^\infty |t-1|e^{-|t|}e^{-st} dt$$

This signal is also non-casual with same reason as part(b) so in general, initial and final value is not valid for bilateral signals.

Question 2

(a) $\left(3e^{-2t} - te^{-2t} + 8e^{-\frac{t}{3}}\right)u(t)$

Poles Calculation:

$$3e^{-2t}, te^{-2t} \rightarrow s_{1,2} = -2, -2 \quad ; \quad 8e^{-\frac{t}{3}} \rightarrow s = -\frac{1}{3}$$

(b) $\left(4e^{-t} \cos(7t + \frac{\pi}{4}) + 9\right)u(t)$

Poles Calculation:

$$\cos\left(7t + \frac{\pi}{4}\right) \rightarrow s_{1,2} = \pm j7 \quad , \quad 9u(t) \rightarrow s = 0$$

Applying the frequency shift property for e^{-t} :

$$e^{-t} \rightarrow -1 \text{ shift} \rightarrow \begin{cases} s_{1,2} = -1 \pm j7 \\ s_3 = 0 \end{cases}$$



Question 3

Delta Function at $t = 0$ Given the system:

$$x(t) = \delta(t) + f(t) \rightarrow X(s) = 1 + \frac{N(s)}{D(s)}$$

Analysis: We have at least four zeros because the degree of the numerator will eventually be greater.

Question 4

One zero at $s = 0$, so the degree of the denominator is greater.

$$x(t) = e^{2t}u(-t) + f(t)u(t)$$

For the left-sided component:

$$\mathcal{L}\{e^{2t}u(-t)\} = -\frac{1}{s-2} \rightarrow p_1 = 2; \text{ROC: } \text{Re}\{s\} < 2$$

Partial Fraction Decomposition:

$$X(s) = \frac{k(s+1)}{(s-2)(s+3)} = k \left(\frac{\frac{3}{5}}{s-2} + \frac{\frac{2}{5}}{s+3} \right)$$

Taking the inverse Laplace transform:

$$\xrightarrow{\mathcal{L}^{-1}} x(t) = -\frac{3k}{5}e^{2t}u(-t) + \frac{2k}{5}e^{-3t}u(t)$$

Question 5

(a) System Function Derivation

Note: For an eigenfunction input $e^{s_0 t}$, the output is $e^{s_0 t}H(s_0)$.

Given differential equation:

$$h'(t) + 2h(t) = (e^{-4t} + b)u(t)$$

Taking the Laplace transform:

$$sH(s) + 2H(s) = \frac{1}{s+4} + \frac{b}{s} \implies H(s) = \frac{b(s+4) + s}{s(s+2)(s+4)}$$

For input $x(t) = \frac{e^{2t}}{12}$, using the eigenfunction property:

$$\frac{e^{2t}}{12} \implies e^{2t}H(2) \implies H(2) = \frac{1}{12} \implies b = \frac{1}{3}$$



(b) Response to DC Input

For input $x_{in}(t) = 1 = e^{0t}$:

$$y(t) = e^{0t}H(0) = \frac{a}{32}$$

Given condition:

$$\frac{a}{32} = \frac{1}{2a} \implies a = \pm 4$$

(c) Filter Types

All-Pass Filter Condition:

$$|H(s)| = 1 \leftrightarrow a = b$$

Minimum Phase Filter Condition: For a stable system (poles < 0), we require zeros < 0 :

$$\text{zeros \& poles } < 0 \rightarrow \frac{-a \pm \sqrt{a^2 - 4}}{2} < 0 \rightarrow -a > \pm \sqrt{a^2 - 4} \rightarrow a > 0$$

Also given $b > 0$, the condition for minimum phase is:

$$a, b > 0$$

Question 6

- (a) The system is causal, so the ROC is right-sided. Since the system $h(t)e^{-5t}$ includes the $j\omega$ axis, it has a Fourier transform.
- (b) The system is causal, so for all $t < 0 \implies h(t) = 0$. Therefore, it is not an odd function.
- (c) The derivative does not affect pole locations. Thus, $\mathcal{L}\{h'(t)\} = sH(s)$ is correct.
- (d) When the pole locations are unaffected and the ROC does not change, the statement is correct.
- (e) Counter-example: $ke^{-t}u(t)$.
- (f) Incorrect because for all $t < 0 \implies h(t) = 0$.

Question 7

(a) Stability Analysis of Open Loop System

Given system function:

$$H(s) = \frac{s - 2}{s^2 + 4s + 4 - a^2}$$



Characteristic equation roots:

$$s^2 + 4s + (4 - a^2) = 0 \implies s_{1,2} = \frac{-4 \pm \sqrt{16 - 4(4 - a^2)}}{2} = \frac{-4 \pm \sqrt{4a^2}}{2} = -2 \pm 2a$$

For a causal and stable LTI system, all poles must be in the left half-plane ($\text{Re}\{s\} < 0$):

$$-2 \pm 2a < 0$$

This gives two inequalities:

$$\begin{cases} -2 - 2a < 0 \implies -2 < 2a \implies a > -1 \\ -2 + 2a < 0 \implies 2a < 2 \implies a < 1 \end{cases} \implies \boxed{-1 < a < 1}$$

(b) Closed Loop Stability Analysis

The closed-loop transfer function with feedback $H_2(s) = b$:

$$G_1(s) = \frac{H_1(s)}{1 + bH_1(s)} = \frac{\frac{s-2}{s^2+4s+4-a^2}}{1 + \frac{b(s-2)}{s^2+4s+4-a^2}} = \frac{s-2}{s^2 + 4s + 4 - a^2 + bs - 2b}$$

Simplifying the denominator:

$$s^2 + (4+b)s + (4-2b-a^2)$$

Setting $a = 0$, the roots are:

$$s_{1,2} = \frac{-(b+4) \pm \sqrt{(b+4)^2 - 4(4-2b)}}{2} = \frac{-b-4 \pm \sqrt{b^2+16b}}{2}$$

For stability (roots < 0):

$$\begin{cases} -b-4 - \sqrt{b^2+16b} < 0 \implies \text{Always true for } b > 0 \\ -b-4 + \sqrt{b^2+16b} < 0 \implies \sqrt{b^2+16b} < b+4 \end{cases}$$

Squaring both sides ($b+4 > 0 \rightarrow b > -4$):

$$b^2 + 16b < b^2 + 8b + 16 \implies 8b < 16 \implies b < 2$$

(c) Stability with $G(s) = \frac{H_1(s)}{(1 + bH_1(s))(1/(s+2)^2)}$

The closed-loop block around $H_1(s)$ is

$$G_1(s) = \frac{H_1(s)}{1 + bH_1(s)} = \frac{s-2}{s^2 + (4+b)s + (4-2b-a^2)}$$



For $a = 0$, this becomes

$$G(s) = \frac{s - 2}{(s^2 + (4 + b)s + (4 - 2b))(s + 2)^2}$$

The poles at $s = -2$ are stable. For the quadratic factor to be stable, its coefficients must be positive:

$$4 + b > 0, \quad 4 - 2b > 0$$

Solving,

$$-4 < b < 2$$

Hence, the system is stable for

$$\boxed{-4 < b < 2}$$